# Xiaotong HE

*University of Paris-Saclay*
*Research in Multimodal AI, Speech Processing and Affective Computing*

*Paris, France*
*+33 7 59 56 98 70*
*xiaotong.he@ieee.org*
*xiaotong-0913*
*xiaotong-he-3a1535364*

## Education

09/2025–09/2026 **Master of Science (M2) in Mechatronics, Computer Vision and AI**, *University of Evry Paris-Saclay*, France, Mention bien

09/2024–07/2025 **Master of Science (M1) in Natural Language Processing (NLP)**, *University of Paris Nanterre, Sorbonne-Nouvelle & Inalco*, France, Mention assez bien

09/2018–07/2022 **Bachelor's Degree in French Language and Linguistics**, *Weifang University*, Weifang, China, (GPA: 3.84/5.00)

## Professional Experience

03/2026 – 09/2026 **Research Intern**, *IBISC Lab, Université Paris-Saclay*, France
- Conducted research on **Multimodal Affective Computing and Emotion Recognition** for Human–Computer/Robot Interaction (HCI/HRI).
- Developed multimodal approaches integrating **speech, audio, EMG, IMU, and motion signals** for emotion recognition and human-state analysis.
- Research resulted in **5 conference papers**, including a **Best Student Paper Award** at WRC SARA 2026.

09/2024–03/2025 **French Corpus Development Intern (French Team Lead)**, *Institute of Language Sciences, Shanghai International Studies University*, Shanghai, China
- Served as French team lead for a French corpus development project under Professor Hu Kaibao.
- Processed 154 French metadata records covering 154,000 tokens, supporting corpus development.

07/2022 – 05/2024 **NLP Engineer / Research Assistant**, *Institute of Bionic Research, Jilin University*, China
- Designed and developed an end-to-end multilingual NLP system based on **LLaMA 3** using **Python** and **Hugging Face**.
- Implemented NLP pipelines including *langdetect*, *spacy*, *jieba*, and data preprocessing scripts.

## Academic & Personal Projects

09/2025 – 02/2026 **Gait Perception and Fatigue Monitoring (AI & Rehabilitation)**
- Designed a multi-sensor system for gait recognition and biomechanical analysis.
- Applied closed-loop control for assistive exoskeletons/simulators.

10/2025 – 11/2025 **Pose Estimation and Robotic Control**, GitHub: xiaotong-0913
- Developed a Python-based system using **OpenPose** for human pose estimation.
- Implemented real-time motion imitation for a simulated robot in **CoppeliaSim**.

11/2024 – 01/2025 **Multilingual Corpus Analysis**, *Project Link: Ouverture*
- Collected and analyzed 166,741 occurrences within a Chinese corpus.
- Developed a web-based co-occurrence analysis system using Bash scripts, Jieba segmentation, and data visualization.

11/2025 **Multimodal Emotion Recognition System**
- Developed a multimodal fusion system (audio, text, pose, facial expressions) using **PyTorch**.

## Skills

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|---|---|
| Programming | **Python**, **Bash, Git, Linux**, HTML/CSS, **SQL**, C/C++ |
| AI & Data | PyTorch, TensorFlow, Hugging Face, OpenCV, Scikit-learn, Pandas, SciPy |
| Robotics | ROS/ROS2, CoppeliaSim/V-REP, MATLAB, SolidWorks |
| Languages | **French** (Fluent, C1), **English** (Professional Working Proficiency, B2), **Chinese** (Native) |

## Core Competencies

Teamwork, Adaptability, Emotional Intelligence, Creativity, Analytical Thinking

## Publications

Springer 2025 *Chen, L., Chen, J., Su, Z., **He, X.**, Zuo, C., & Li, X. (2025). A Vision-Based AI framework for skill transfer and robotic replication of Miao embroidery techniques. Signal, Image and Video Processing, 19(9), 769.*

MDPI 2025 *Shen J, Chen L, **He X**, et al. An Interactive Human-in-the-Loop Framework for Skeleton-Based Posture Recognition in Model Education[J]. Biomimetics, 2025, 10(7): 431.*

## Conference Papers

IEEE 2026 ***He, X.**, Hu, W., Wang, F., Schmirander, Y., Su, H., Dychus, E., & Alfayad, S. (2026). SongEmotionTrack: Affective Telepresence for Remote Music Interaction with Lyrics, Singing Voice, and Melody. 2026 IEEE Conference on Telepresence, Bristol, United Kingdom.*

WRC SARA 2026 ***He, X.**, Hu, W., Wang, F., Schmirander, Y., Su, H., Dychus, E., & Alfayad, S. (2026). Rehab-EmotionTrack: Toward Patient-Role-Aware Speech Evidence Aggregation for Therapist-Facing Rehabilitation State Monitoring. 8th WRC Symposium on Advanced Robotics and Automation (WRC SARA 2026), Beijing, China.*

IEEE 2026 *Hu, W., **He, X.**, Wang, F., Schmirander, Y., Su, H., Dychus, E., & Alfayad, S. (2026). Towards Human-in-the-Loop Wearable Cross-Modal Sensing for Physiological Feedback in Piano Telepresence Training. 2026 IEEE Conference on Telepresence, Bristol, United Kingdom.*

WRC SARA 2026 *Hu, W., **He, X.**, Wang, F., Schmirander, Y., Su, H., Dychus, E., Alfayad, S., & Couprie, P. (2026). A Novel Audio-Supervised Cross-Modal Framework with Wearable EMG-IMU Sensing for Emotion-Aware Expressive Music Performance Analysis. 8th WRC Symposium on Advanced Robotics and Automation (WRC SARA 2026), Beijing, China.* **[Best Student Paper Award]**

IEEE ARM 2026 *Xing, Z., **He, X.**, Hu, W., Feng, B., Wang, F., Schmirander, Y., Su, H., & Alfayad, S. (2026). Parameterized Symmetry and Fuzzy Modulation for Unilateral Hip Assistance in a Lower-Limb Exoskeleton. IEEE ARM 2026, Paris, France.*

## Awards

2026 Best Student Paper Award, *2026 WRC Symposium on Advanced Robotics and Automation (WRC SARA)*

## Academic Training

06/2026 – 07/2026 HI! PARIS Summer School, *Télécom Paris*